



MULTIMEDIA UNIVERSITY OF KENYA
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2020/2021

THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE /BACHELOR OF SCIENCE IN
SOFTWARE ENGINEERING

CCS 2315: DISTRIBUTED SYSTEMS

DATE: 20TH AUGUST, 2021

TIME: 2 HOURS

INSTRUCTIONS:

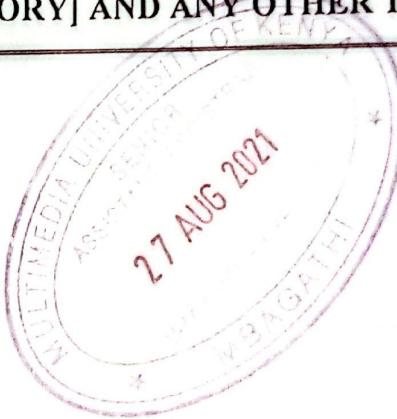
ANSWER YOUR QUESTIONS IN THE ANSWER BOOKLET PROVIDED
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE [THIRTY MARKS]

a) Define the following terms.

[4 Marks]

- i. Distributed systems.
- ii. Network operating system.
- iii. Inter-process Communication.
- iv. Protocol



b) Explain the two ways a Distributed File System can be implemented.

[4 Marks]

c) State and explain the three capacity configurations of queues that are used to pass messages in a link.

[6 Marks]

d) State and explain the two types of Network operating systems.

[4 Marks]

e) State the advantages of Remote Procedure Call.

[5 Marks]

f) State and explain the two ways clock synchronization is achieved in distributed systems.

[4 Marks]

g) Discuss the Bully Election Algorithm in Distributed processing.

[3 Marks]

QUESTION TWO [TWENTY MARKS]

- a) Differentiate between direct communication links and in-direct communication links. [2 Marks]
- b) State and explain the two functions of inter-process communication. [4 Marks]
- c) Explain the three kinds of problems that can occur in a distributed system. [6 Marks]
- d) Discuss the four basic architecture models of distributed systems. [8 Marks]

QUESTION THREE [TWENTY MARKS]

- a) Explain the two components of a Distributed File System. [4 Marks]
- b) Discuss the five main characteristics of inter-process communication in a distributed environment/system. [10 Marks]
- c) State and explain the three approaches based on message passing that are used to implement mutual exclusion in distributed systems. [6 Marks]

QUESTION FOUR [TWENTY MARKS]

- a) Discuss the requirements of a mutual exclusion algorithm. [8 Marks]
- b) Discuss the methods implemented by Inter-process communication (IPC) to allow running of programs concurrently. [12 Marks]

QUESTION FIVE [TWENTY MARKS]

- a) Using a diagram, list the steps that take place during a Remote Procedure Call. [8 Marks]
- b) Giving examples of two processes want to communicate with each other, discuss the two ways processes can communicate with each other. [6 Marks]
- c) Discuss the types of group communication in a distributed system. [6 Marks]



MULTIMEDIA UNIVERSITY OF KENYA
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UNIVERSITY EXAMINATIONS 2023/2024

THIRD YEAR SUPPLEMENTARY/SPECIAL EXAMINATION FOR THE DEGREE OF
BACHELOR OF BUSINESS INFORMATION TECHNOLOGY

BIT 2323: DISTRIBUTED SYSTEMS

DATE: 31ST JULY 2024

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN THE ANSWER BOOKLET PROVIDED

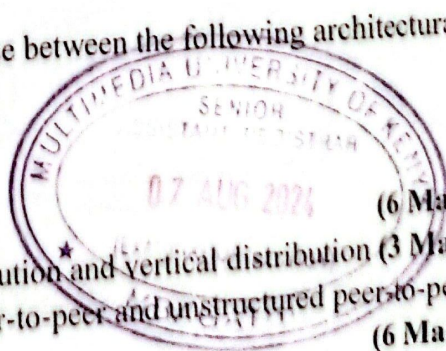
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) Provide two distinct definitions of the term distributed system (2 Marks)
- b) Explain what is distribution transparency and distinguish the difference between the following types of transparency
 - 1) Location Transparency (4 Marks)
 - 2) Migration Transparency (5 Marks)
- c) With the help of a figure describe the management of nodes in a chord system (4 Marks)
- d) What is a three tiered client server architecture (5 Marks)
- e) Apart from improved reliability, what are four other reasons that motivate organizations to procure distributed systems? (5 Marks)
- f) Describe how a coordinator can be elected using a Ring Algorithm (5 Marks)
- g) Describe how the Berkeley UNIX algorithm is used in time servers (5 Marks)

QUESTION TWO (TWENTY MARKS)

- a) Using relevant diagrams distinguish the difference between the following architectural configuration in distributed systems
 - 1) Shared Data Space (6 Marks)
 - 2) Object-Based Architecture
 - 3) Event-Based Architecture
- b) What is the difference between horizontal distribution and vertical distribution (3 Marks)
- c) Describe the difference between a structured peer-to-peer and unstructured peer-to-peer architecture (6 Marks)



- d) Describe five different characteristics of a distributed system

QUESTION THREE (TWENTY MARKS)

- Explain why a receiver call cannot be asynchronous
- With help of figures differentiate between the following
 - An Asynchronous Execution
 - A Synchronous Execution
- Describe the concept behind Remote Procedure Call (RPC)
- Using relevant diagrams explain how the following procedure calls function
 - A single Machine Procedure Call
 - RPC in a Client and Server Stub

QUESTION FOUR (TWENTY MARKS)

- Name three different types of distributed systems
- Distinguish the main differences between cluster computing systems and grid computing systems
- With the help of a diagram describe the operation of a cluster computing system
- Differentiate between the following layers of a grid computing system
 - Fabric Layer
 - Connectivity Layer
 - Resource Layer
 - Collective Layer

QUESTION FIVE (TWENTY MARKS)

- With the help of a diagram explain how a simplified Internet search engine is implemented using a three tier architecture and its operation
- What is a super peer in a peer to peer network and what reasonable requirements should a super peer meet
- With the aid of a diagram explain the role of middleware in a distributed system
- Using Figure (1) below, Explain what happens when Coordinator No 7 has crashed and process No 2 and No 5 build election messages

